

Rope Set

Technical Datasheet - Generic Template

Purpose: This document provides a generic technical information template for a theatre rope set or line set. Complete the blank fields with manufacturer, installer and site-specific information before relying on it as an equipment record.

1. Equipment overview

A **rope set** is a manual rope-operated lifting or suspension arrangement used in theatre and entertainment environments to raise, lower or hold bars, scenery, curtains, cloths or other approved production loads. The rope set should be assessed as a complete system, including ropes, pulleys or sheaves, head blocks, loft blocks, operating lines, cleats or locking devices, load bars, terminations, guides and the supporting structure.

- The rope set must only be used within its stated Safe Working Load / Working Load Limit and any site-specific restrictions.
- The complete load path should be understood, including the load, bar, attachment points, ropes, blocks, terminations and structure.
- Rope sets should only be operated by competent persons who understand the system, travel limits, balancing method and local operating procedure.
- Any change of load, travel, rope type, termination method or support arrangement should be assessed before use.

Identification and technical data

Equipment name	Rope Set
Asset / reference number	To be completed
Venue / room / location	To be completed
Associated bar / batten / load	To be completed
Manufacturer / installer	To be completed
Date installed / commissioned	To be completed
SWL / WLL	To be confirmed from design, marking or competent-person record
Rope type and diameter	To be completed
Number of lift lines	To be completed
Travel / operating range	To be completed
Operating method	Hand line / cleat / brake / other - confirm site details
Inspection interval	To be confirmed by competent person / written scheme / site procedure

2. Applicable regulatory context

This section is a guidance summary. It does not replace manufacturer instructions, engineering design information, risk assessment, a competent-person examination report or a written scheme of examination.

Regulations and guidance

LOLER 1998	Relevant where the rope set is lifting equipment or forms part of a lifting arrangement. Key duties include strength and stability, suitable marking, planning of lifting operations and thorough examination by a competent person where required.
PUWER 1998	Relevant because the rope set is work equipment. It must be suitable for the intended use, maintained in a safe condition, inspected where deterioration could create risk, and used by people with adequate information, instruction and training.
BS EN 17206	Relevant where the rope set forms part of entertainment machinery or a stage lifting system. Consider load-bearing components, prevention of uncontrolled movement, inspection, testing and documentation.
ABTT Technical Standards	Useful theatre-industry good practice for inspection, maintenance, operational control and record keeping within places of entertainment.

3. System components to identify

- Load bar, batten or other suspended item connected to the rope set.
- Rope or wire rope lift lines, including type, diameter, construction and condition.
- Head blocks, loft blocks, diverter pulleys, sheaves and any return pulleys.
- Rope terminations, thimbles, clamps, ferrules, knots or approved termination devices.
- Operating line, cleat, brake, locking device or other load-holding arrangement.
- Support structure, beam, grid, fixing points, brackets and any local strengthening.
- Travel path, stops, guides and zones where the load or lines may snag or foul.
- Identification labels, SWL/WLL markings, inspection tags and record references.

4. Safe use and limitations

- Do not exceed the confirmed SWL/WLL of the rope set or any component in the load path.
- Do not use the system if the load path, rope condition, termination method or rating is uncertain.
- Do not shock load, side load or drag loads unless the system has been specifically designed for that use.
- Do not alter rope routing, replace rope, change terminations or modify supports without competent assessment.
- Avoid trapping, abrasion, twisting, kinking or contamination of ropes and wire ropes.
- Keep the travel path clear and check that loads cannot foul scenery, structure, services, lighting equipment or people.
- Follow the venue operating procedure for loading, balancing, securing and controlling the set.

5. Pre-use and periodic inspection checklist

Check item	Pass / fail / N/A	Notes / action required
Identification, SWL/WLL and operating limits are known and legible.	☐ ☐ ☐	
Rope set corresponds with asset records, drawings or site register.	☐ ☐ ☐	
Ropes / wire ropes are free from broken wires, fraying, cuts, kinks, crushing, flattening, corrosion or chemical damage.	☐ ☐ ☐	
Rope diameter, type and construction match the system design and sheave grooves.	☐ ☐ ☐	
Rope terminations are correctly fitted, secure and free from slippage or distortion.	☐ ☐ ☐	
Head blocks, loft blocks and pulleys are secure, correctly aligned and free running.	☐ ☐ ☐	
Sheaves are not cracked, excessively worn, seized, sharp edged or incorrectly grooved.	☐ ☐ ☐	
Anchors, brackets, bolts, clamps and supports are secure and free from movement, corrosion or damage.	☐ ☐ ☐	
Operating line, cleat, brake, lock or holding device is effective and suitable for the load.	☐ ☐ ☐	
Load bar, batten or attached item is secure, straight and free from signs of overload.	☐ ☐ ☐	
Travel path is clear and the set moves smoothly without snatching, rubbing, jumping or uncontrolled movement.	☐ ☐ ☐	
Records of inspection, maintenance and thorough examination are available and up to date.	☐ ☐ ☐	

6. Defects requiring action

Remove from service, isolate or seek competent-person advice if any safety-critical defect is found or if the rating,

installation details or condition is uncertain.

- Missing or unreadable SWL/WLL information where the capacity cannot be otherwise verified.
- Broken wires, severe corrosion, heavy fraying, damaged fibres, crushed sections, kinks or flattened rope.
- Loose, slipping, incorrectly fitted or non-standard terminations.
- Cracked, seized, misaligned or excessively worn pulleys, sheaves or blocks.
- Loose structural fixings, cracked welds, distorted brackets or movement in the support structure.
- Failure of a cleat, brake, lock, rope lock or other holding arrangement to secure the load.
- Uncontrolled descent, drift, snatching, abnormal noise, excessive friction or rubbing.
- Evidence of unauthorised modification, non-standard repair, changed rope routing or unsuitable replacement parts.

7. Maintenance and documentation

- Maintain the system in accordance with manufacturer instructions, installer information and the venue inspection regime.
- Keep inspection and thorough examination reports with the equipment record.
- Record rope changes, replacement parts, lubrication, adjustment, repairs and any change in use.
- After significant repair, alteration, overload, impact or change in configuration, obtain competent-person assessment before returning the set to service.
- Store related documents such as drawings, certificates, load test records and written scheme references in a controlled location.

Inspection record summary

Last formal inspection / thorough examination	To be completed
Competent person / organisation	To be completed
Report reference	To be completed
Next inspection due	To be completed
Defects recorded	Yes / No - details to be completed
Returned to service by	To be completed

8. Site-specific notes

Reference note

Regulatory context in this template has been aligned with HSE guidance on LOLER and PUWER. HSE guidance states that lifting equipment should be fit for purpose, suitable for the task, suitably marked and, in many cases, subject to statutory periodic thorough examination. PUWER guidance also addresses suitability, maintenance, inspection and training for work equipment.